MD. Mehedi Hasan Maruf

Software Engineer

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Summary

Computer Science engineer (RUET) with a B.Sc. thesis on multi-modal curriculum bottleneck detection, fusing prerequisite-graph centrality, Bayesian fail-rate modeling, and aspect-based sentiment analysis into a single bottleneck score validated causally (DoWhy) on 23,000+ real student records. Also has almost 4 years of industry experience building production AI systems, including a LangChain-based generative AI agent platform and RAG pipelines, plus full-stack TypeScript/Python backend development and DevOps work (Docker, CI/CD) at Aamira Software Solutions.

Skills

Programming Languages: TypeScript, Python, JavaScript
Web Technologies: Angular, React.js, Next.js, Node.js, FastAPI, Django
Databases: MySQL, PostgreSQL, MongoDB, Supabase
Tools & Concepts: Git, Docker, CI/CD (GitHub Actions, Netlify), VPS (Linux) Management, REST APIs, ORM (Django ORM, Supabase), Project Management, Problem Solving
AI/ML: LangChain, Semantic Analysis, NLP, RAG, Causal Inference (DoWhy), Bayesian Modeling, Graph Neural Networks (PyTorch)

Experience

Software Engineer II — *Jan 2025 – Present*

Aamira Software Solutions, Kokomo, Indiana, USA

- Led development of a customer relationship management system that increased client response rates by 30%.
- Architected and implemented a Typescript microservice backend solution that reduced API response time by 40%.
- Optimized database queries using Supabase, resulting in 60% improvement in application performance.

Industrial Trainee — *Feb 2025 – Mar 2025*

Brain Station 23, Dhaka, Bangladesh

- Gained practical experience in enterprise software development methodology and practices.
- Served as Team Representative for a 15-person cohort, coordinating communication between management and trainees.
- Contributed to a student AI system using Next.js and Node.js that was deployed to an online server.

Junior Software Engineer — *Nov 2022 – Apr 2024*

Code Studio, Rajshahi, Bangladesh

- Developed backends supporting thousands of daily users using JavaScript, TypeScript, and Python (Django).
- Managed 12 VPS (Linux) with 99.8% uptime and optimized CI/CD pipelines, reducing deployment time by 40%.
- Implemented Docker containerization, simplifying deployment across 5 different client environments.

Personal Projects

StudentAI - Intelligent Learning Companion — *Feb 2025 – Mar 2025*

Read More:

- Developed during industrial attachment at Brain Station 23: AI-powered learning assistant combining **FastAPI backend** with **Next.js frontend** for personalized education.
- Implemented a **RAG pipeline** with LangChain for document Q&A, semantic retrieval, and conversation memory; extended to YouTube content analysis and intelligent quiz generation.
- Built a **multimodal** document processing pipeline supporting PDFs, images, and videos with real-time WebSocket chat, JWT authentication, and progress tracking.
- Deployed production-ready application on Vercel (frontend) and Render (backend) with Supabase database integration.

Omnious – Visual Debugging Platform — *Nov 2025 – Present*

Read More:

- Building a full-stack visual debugging platform with interactive code-graph visualization, real-time trace animation, error heatmaps, and **AI-powered** chat-based debugging.

- Engineered a scalable **Node.js** backend using **Fastify 5** and **tRPC v11** with **TypeScript**, integrated with a **Next.js 16/React 19** frontend in a **Turborepo** monorepo.
- Integrated **PostgreSQL 15** with **pgvector**, **ltree**, and **pg_trgm** via **Supabase** for semantic search and hierarchical data; implemented Row-Level Security (RLS).
- Containerized the full-stack application using **Docker Compose** with multi-stage builds; automated deployment via **GitHub Actions** CI/CD pipeline.

Research

TRACE: A Multi-Modal Framework for Curriculum Bottleneck Detection and Causal Validation

Jul 2025 – Aug 2026

B.Sc. Thesis, Dept. of CSE, RUET | Supervisor: Farjana Parvin, Assistant Professor

- **Architected** TRACE, a multi-modal graph-mining framework fusing prerequisite-graph centrality (NetworkX, GAT edge weighting), Bayesian fail-rate modeling, and aspect-based sentiment analysis (DeBERTa/VADER) into a single composite bottleneck score.
- **Built** a reproducible synthetic-curriculum benchmark with planted, type-disjoint bottlenecks; fusion reached an F-measure of 0.75 and NDCG@10 of 0.81, roughly double the best single-signal baseline.
- **Causally validated** flagged bottlenecks on OULAD (23,369 students) via **DoWhy** backdoor-adjustment OLS; high assignment engagement raised held-out pass rates by 19.9 percentage points ($p < 0.01$, E-value 1.89).
- **Demonstrated** that the structure-only detector generalizes across RUET, MIT, and Stanford curricula (top-5 hub courses gate 44–55% of the downstream curriculum) and validated rankings against a domain expert (8/10 agreement); surfaced RUET's high-failure MATH1213 (16.9% fail rate) via the "Preparation Paradox," where student effort masks true course difficulty.

Education

B.Sc. Engineering, Computer Science and Engineering

Mar 2022 – Aug 2026

Rajshahi University of Engineering & Technology (RUET), Rajshahi, Bangladesh

- CGPA: 3.75

Certifications & Achievements

- Digital Accessibility Training (DAT) - a2i (Aspire to Innovate), Govt. of Bangladesh (Oct 2025)
- Industrial Training Certificate - Brain Station 23 (Mar 2025)
- PHP (Laravel) - EDGE Project (Feb 2025)
- Experience Certificate - Code Studio (Apr 2024)
- 4th Place - RUET CodeSmash (Team Competition) - Competitive Programming Contest (Mar 2023)
- 4th Place - RUET CodeSpark (Individual) - Competitive Programming Contest (Oct 2022)